THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The SI unit of electric charge is 1

(a) coulomb **(b)** ampere **(c)** volt **(d)** ohm

2. Ohm's law states that the current through a conductor is 1

(a) directly proportional to the potential difference, at constant temperature

(b) independent of the potential difference

(c) inversely proportional to the potential difference

(d) proportional to the square of V

3. For resistors in series, the equivalent resistance is 1

(a) 1/R1 + 1/R2 **(b)** R1 + R2 + R3

(c) less than each **(d)** the average of the resistances

4. The SI unit of electric current is 1

(a) watt **(b)** ampere **(c)** volt **(d)** coulomb

5. The SI unit of resistance is 1

(a) watt **(b)** volt **(c)** ampere **(d)** ohm

6. In a series circuit the quantity that is the same through all the resistors is 1

(a) power **(b)** current

(c) potential difference **(d)** resistance

7. Electric current is measured by an 1

(a) voltmeter connected in series **(b)** ammeter connected in parallel

(c) voltmeter connected in parallel **(d)** ammeter connected in series

8. The V-I graph for an ohmic conductor is 1

(a) a horizontal line **(b)** a straight line through the origin

(c) a curve **(d)** a parabola

9. In a parallel circuit the quantity that is the same across all the resistors is 1

(a) resistance **(b)** power

(c) potential difference **(d)** current

10. Potential difference is measured by a 1

(a) galvanometer in series **(b)** voltmeter connected in series

(c) ammeter in parallel **(d)** voltmeter connected in parallel

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Define one ampere of current. 2

12. List the factors on which the resistance of a conductor depends. 2

13. Write the expressions for power in three forms. 2

14. What is meant by the electric potential difference between two points? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A current of 0.5 A flows in a conductor for 10 minutes. Find the charge that flows and the number of electrons. 3

16. Find the equivalent resistance when 2 ohm, 3 ohm and 5 ohm are connected in series, and the current from a 6 V battery. 3

17. Why does a bulb filament glow but the connecting wire does not? 3

OR

How much work is done in moving a charge of 2 C across two points having a potential difference of 12 V? 3

18. Three resistors 20 ohm, 30 ohm and 60 ohm are in parallel. Find the equivalent resistance. 3

OR

How much energy in kWh is consumed by a 2 kW heater in 3 hours? What is the cost at Rs 5 per unit? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. State and explain Ohm's law. Describe an activity to verify it and how the V-I graph is used to find the resistance. 5

OR

Establish the formula H = I^2 R t from the definitions of work and potential difference. 5

20. Derive the formula for the equivalent resistance of three resistors connected in series. Find it for 3 ohm, 4 ohm and 5 ohm. 5

OR

List five applications of the heating effect of electric current with the principle of working of a fuse. 5

END OF QUESTION PAPER — Total: 40 Marks